# Improper Integrals

## 7.8

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# Improper Integrals

In this section, we will learn:

How to solve definite integrals

where the interval is infinite and

where the function has an infinite discontinuity.

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# IMPROPER INTEGRALS

- **In this section, we extend the concept of a definite integral to the cases where:**
  - The interval is infinite
  - *f* has an infinite discontinuity in [*a*, *b*]

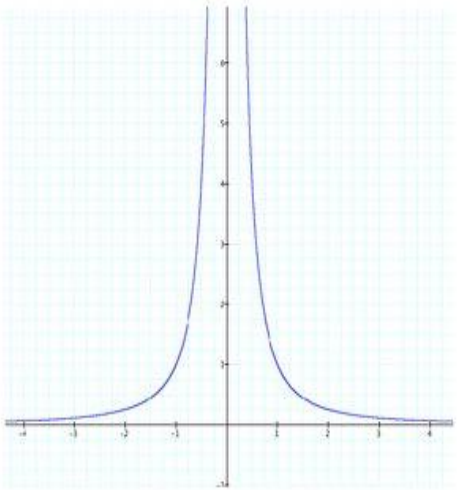
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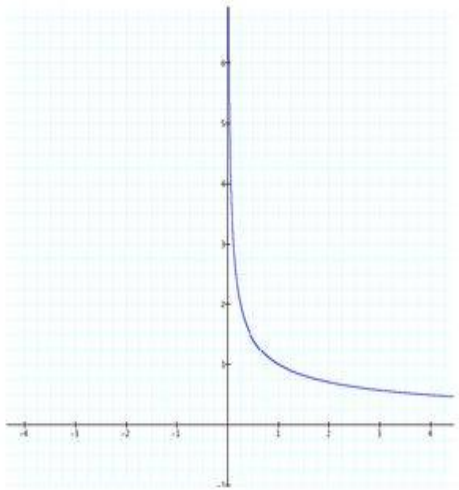
- **Suppose *f* is a positive continuous function defined on a finite interval but has a vertical asymptote at a point inside the given interval.**

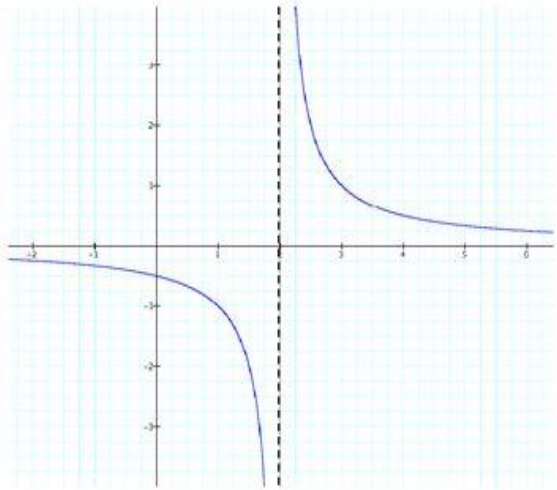
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# Vertical Asymptote

If $f(x)$ approaches infinity (or negative infinity) as $x$ approaches $c$ from the right or the left, then the line $x = c$ is a vertical asymptote of the graph of $f$.







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# Definition

If $f$ is continuous on $[a, b)$ and is discontinuous at $b$, then

$$\int_a^b f(x)\,dx = \lim_{t \to b^-} \int_a^t f(x)\,dx$$

if this limit exists (as a finite number).

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# Definition

If *f* is continuous on (*a*, *b*] and is discontinuous at *a*, then

$$\int_a^b f(x)\,dx = \lim_{t \to a^+} \int_t^b f(x)\,dx$$

if this limit exists (as a finite number).

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# Definition

If $f$ has a discontinuity at $c$, where $a < c < b$, and both $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ are convergent, then we define:

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

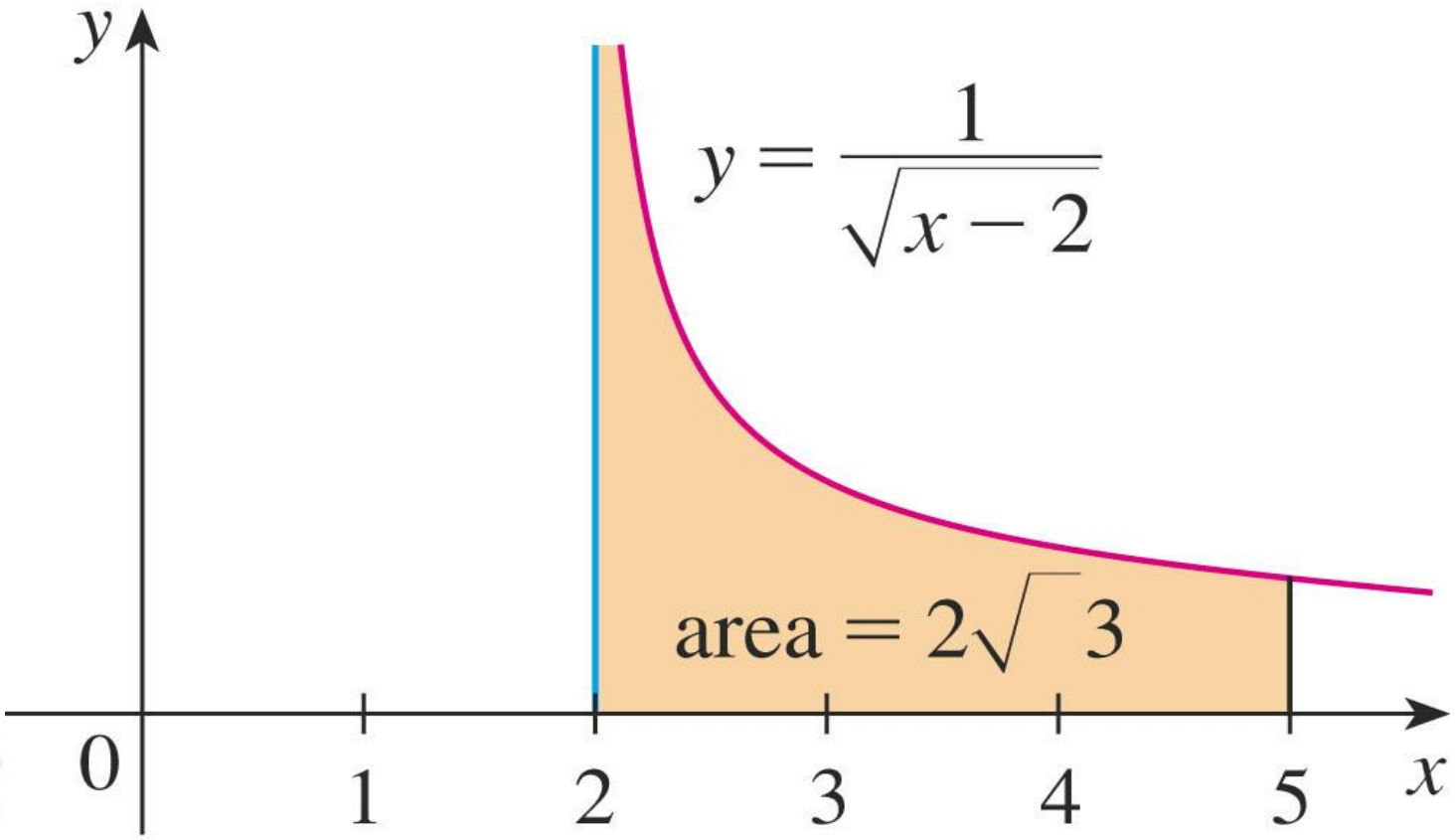
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# Example

Find $\int_2^5 \frac{1}{\sqrt{x-2}}\,dx$

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# Remark




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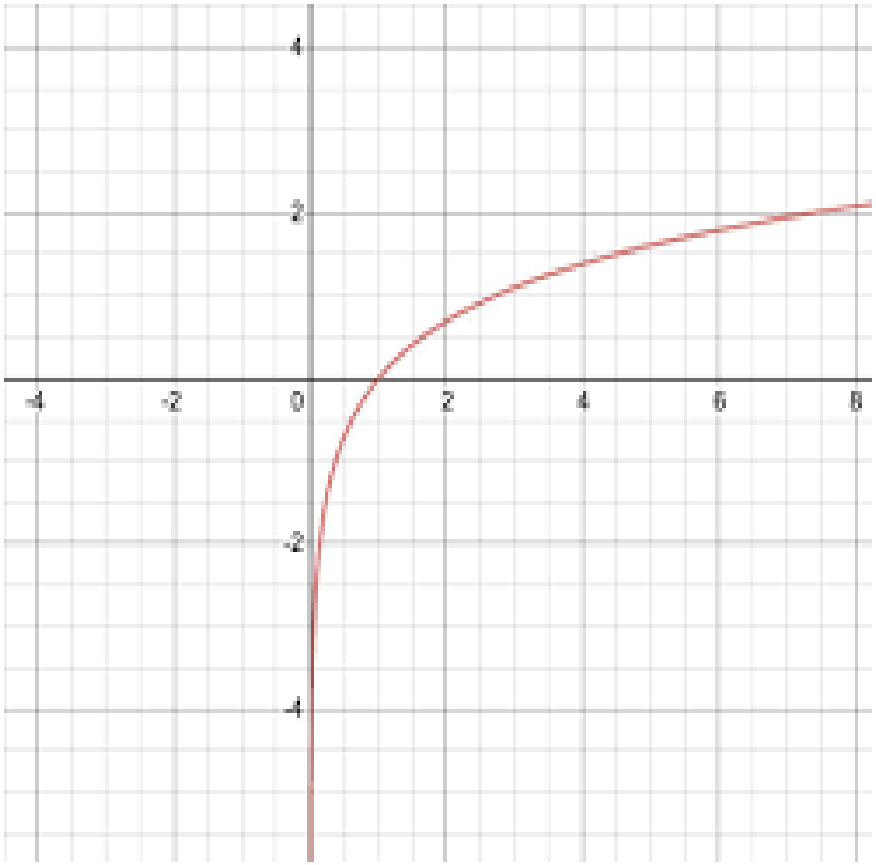
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# Example

Evaluate $\int_0^3 \frac{dx}{x-1}$ if possible.

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# Example:




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# Example

Determine whether $\int_0^{\pi/2} \sec x \, dx$ converges or diverges.

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# WARNING

- **Suppose we had not noticed the asymptote in last example and had, instead, confused the integral with an ordinary integral.**

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# WARNING

- **Then, we might have made the following erroneous calculation:**

$$\int_0^3 \frac{dx}{x-1} = \ln|x-1|\Big]_0^3$$

$$= \ln 2 - \ln 1$$

$$= \ln 2$$

  - **This is wrong because the integral is improper and must be calculated in terms of limits.**

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# L'Hospital Rule

If $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{0}{0}$ or $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\infty}{\infty}$, then $\lim_{x \to a} \frac{f'(x)}{g'(x)}$

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# Applications

- One of the most important applications of this idea, probability distributions which is a mathematical function that gives the probabilities of occurrence of different possible outcomes for an experiment, will be studied in statistic

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# Definition

If $\int_a^t f(x)\,dx$ exists for every number $t \geq a$, then

$$\int_a^{\infty} f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx$$

provided this limit exists (as a finite number).

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# Definition

If $\int_t^b f(x)\,dx$ exists for every number $t \le a$, then

$$\int_{-\infty}^{b} f(x)\,dx = \lim_{t \to \infty} \int_t^b f(x)\,dx$$

provided this limit exists (as a finite number).

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# Remark:

The improper integrals $\int_a^{\infty} f(x)\,dx$ and $\int_{-\infty}^{b} f(x)\,dx$ are called:

- Convergent if the corresponding limit exists.
- Divergent if the limit does not exist.

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## Definition 1

- If both $\int_a^{\infty} f(x)\,dx$ and $\int_{-\infty}^{a} f(x)\,dx$ are convergent, then we define:

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{a} f(x)\,dx + \int_a^{\infty} f(x)\,dx$$

  - Here, any real number *a* can be used.

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# Example:

- Consider the infinite region *S* that lies:
  - Under the curve $y = 1/x^2$
  - Above the *x*-axis
  - To the right of the line $x = 1$

Then Find

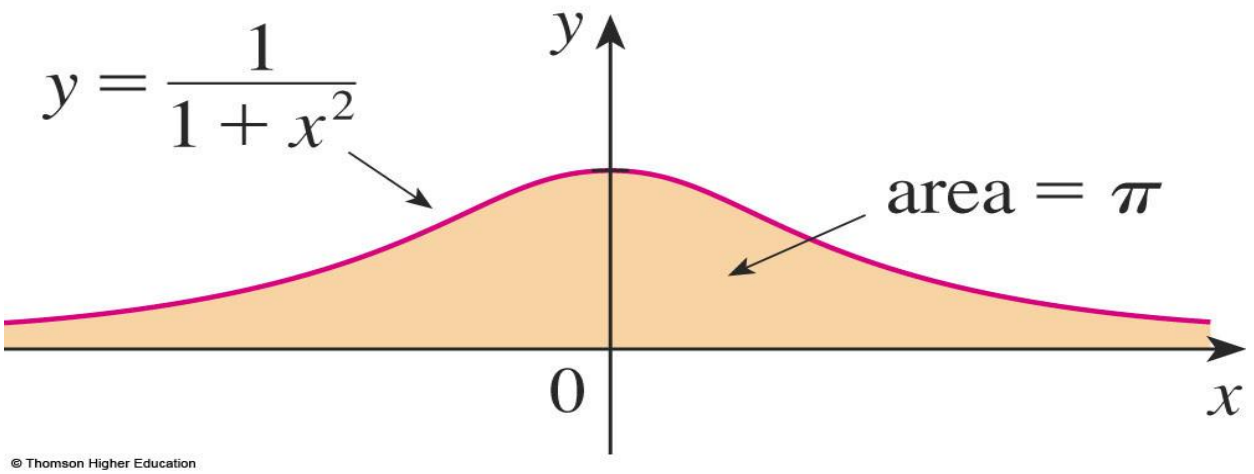
$$\int_1^\infty \frac{1}{x^2}\,dx$$

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## Example 3

- Evaluate

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx$$




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## Homework

- Determine whether the integral

$$\int_1^{\infty} (1/x)\,dx$$

is convergent or divergent.

- Evaluate $\int_{-\infty}^{0} xe^{x}\,dx$

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